

Experiment-8

Customer Churn Prediction using Decision Tree Classification

```
In [1]: # Step 1: Load Dataset

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    confusion_matrix,
    classification_report
)
from sklearn.tree import plot_tree

df = pd.read_csv("WA_Fn-UseC_-Telco-Customer-Churn.csv")

print("First 5 Records:")
print(df.head())

# Step 2: Explore Dataset

print("\nDataset Shape:")
print(df.shape)

print("\nDataset Information:")
print(df.info())

print("\nStatistical Summary:")
print(df.describe())

print("\nMissing Values:")
print(df.isnull().sum())

print("\nDuplicate Values:")
print(df.duplicated().sum())

# Step 3: Data Preprocessing

df = df.drop_duplicates()

df["TotalCharges"] = pd.to_numeric(
    df["TotalCharges"],
    errors="coerce"
)

df["TotalCharges"] = df["TotalCharges"].fillna(
```

```
        df["TotalCharges"].median()
    )

    print("\nMissing Values After Preprocessing:")
    print(df.isnull().sum())

    # Step 4: Remove Unnecessary Columns

    df = df.drop(
        columns=["customerID"]
    )

    print("\nColumns After Removing Customer ID:")
    print(df.columns)

    # Step 5: Encode Categorical Variables

    categorical_columns = df.select_dtypes(
        include=["object"]
    ).columns

    encoder = LabelEncoder()

    for column in categorical_columns:
        df[column] = encoder.fit_transform(
            df[column]
        )

    print("\nEncoded Dataset:")
    print(df.head())

    # Step 6: Select Features and Target

    X = df.drop(
        columns=["Churn"]
    )

    y = df["Churn"]

    print("\nFeatures:")
    print(X.columns)

    print("\nTarget:")
    print(y.name)

    # Step 7: Split Dataset into Training and Testing Data

    X_train, X_test, y_train, y_test = train_test_split(
        X,
        y,
        test_size=0.20,
        random_state=42,
        stratify=y
    )

    print("\nTraining Data Shape:")
```

```
print(X_train.shape)

print("\nTesting Data Shape:")
print(X_test.shape)

# Step 8: Train Decision Tree Classifier

model = DecisionTreeClassifier(
    criterion="gini",
    max_depth=5,
    random_state=42
)

model.fit(
    X_train,
    y_train
)

print("\nDecision Tree Model Trained Successfully.")

# Step 9: Predict Customer Churn

y_pred = model.predict(
    X_test
)

print("\nPredicted Values:")
print(y_pred[:20])

# Step 10: Evaluate Model

accuracy = accuracy_score(
    y_test,
    y_pred
)

precision = precision_score(
    y_test,
    y_pred
)

recall = recall_score(
    y_test,
    y_pred
)

f1 = f1_score(
    y_test,
    y_pred
)

print("\nModel Evaluation:")

print("Accuracy :", round(accuracy, 4))
print("Precision:", round(precision, 4))
print("Recall   :", round(recall, 4))
print("F1 Score :", round(f1, 4))
```

```
# Step 11: Classification Report

print("\nClassification Report:")

print(
    classification_report(
        y_test,
        y_pred
    )
)

# Step 12: Confusion Matrix

cm = confusion_matrix(
    y_test,
    y_pred
)

print("\nConfusion Matrix:")
print(cm)

plt.figure(figsize=(7, 5))

sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=["No Churn", "Churn"],
    yticklabels=["No Churn", "Churn"]
)

plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")

plt.show()

# Step 13: Visualize Decision Tree

plt.figure(figsize=(20, 10))

plot_tree(
    model,
    feature_names=X.columns,
    class_names=["No Churn", "Churn"],
    filled=True,
    rounded=True,
    fontsize=8
)

plt.title("Decision Tree for Customer Churn Prediction")

plt.show()
```

```
# Step 14: Find Feature Importance

feature_importance = pd.DataFrame({
    "Feature": X.columns,
    "Importance": model.feature_importances_
})

feature_importance = feature_importance.sort_values(
    by="Importance",
    ascending=False
)

print("\nFeature Importance:")
print(feature_importance)

# Step 15: Visualize Feature Importance

plt.figure(figsize=(10, 6))

sns.barplot(
    data=feature_importance,
    x="Importance",
    y="Feature"
)

plt.title("Feature Importance in Customer Churn Prediction")
plt.xlabel("Importance")
plt.ylabel("Feature")

plt.show()

# Step 16: Analyze Churn Distribution

churn_count = df["Churn"].value_counts()

print("\nCustomer Churn Distribution:")
print(churn_count)

plt.figure(figsize=(6, 5))

sns.countplot(
    x=df["Churn"]
)

plt.title("Customer Churn Distribution")
plt.xlabel("Churn")
plt.ylabel("Number of Customers")

plt.show()

# Step 17: Display Customers Predicted to Churn

churn_predictions = X_test.copy()

churn_predictions["Actual Churn"] = y_test.values
churn_predictions["Predicted Churn"] = y_pred
```

```
print("\nCustomers Predicted to Churn:")

print(
    churn_predictions[
        churn_predictions["Predicted Churn"] == 1
    ].head(20)
)

# Step 18: Final Result

print("\nFinal Result:")

print(
    "Decision Tree Customer Churn Prediction completed successfully."
)

print(
    f"Model Accuracy: {accuracy * 100:.2f}%"
)

print(
    f"Precision: {precision * 100:.2f}%"
)

print(
    f"Recall: {recall * 100:.2f}%"
)

print(
    f"F1 Score: {f1 * 100:.2f}%"
)
```

First 5 Records:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	\
0	7590-VHVEG	Female	0	Yes	No	1	No	
1	5575-GNVDE	Male	0	No	No	34	Yes	
2	3668-QPYBK	Male	0	No	No	2	Yes	
3	7795-CFOCW	Male	0	No	No	45	No	
4	9237-HQITU	Female	0	No	No	2	Yes	

	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	\
0	No phone service	DSL	No	...	No	
1	No	DSL	Yes	...	Yes	
2	No	DSL	Yes	...	No	
3	No phone service	DSL	Yes	...	Yes	
4	No	Fiber optic	No	...	No	

	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	\
0	No	No	No	Month-to-month	Yes	
1	No	No	No	One year	No	
2	No	No	No	Month-to-month	Yes	
3	Yes	No	No	One year	No	
4	No	No	No	Month-to-month	Yes	

	PaymentMethod	MonthlyCharges	TotalCharges	Churn
0	Electronic check	29.85	29.85	No
1	Mailed check	56.95	1889.5	No
2	Mailed check	53.85	108.15	Yes
3	Bank transfer (automatic)	42.30	1840.75	No
4	Electronic check	70.70	151.65	Yes

[5 rows x 21 columns]

Dataset Shape:
(7043, 21)

Dataset Information:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 7043 entries, 0 to 7042

Data columns (total 21 columns):

#	Column	Non-Null Count	Dtype
0	customerID	7043 non-null	object
1	gender	7043 non-null	object
2	SeniorCitizen	7043 non-null	int64
3	Partner	7043 non-null	object
4	Dependents	7043 non-null	object
5	tenure	7043 non-null	int64
6	PhoneService	7043 non-null	object
7	MultipleLines	7043 non-null	object
8	InternetService	7043 non-null	object
9	OnlineSecurity	7043 non-null	object
10	OnlineBackup	7043 non-null	object
11	DeviceProtection	7043 non-null	object
12	TechSupport	7043 non-null	object
13	StreamingTV	7043 non-null	object
14	StreamingMovies	7043 non-null	object
15	Contract	7043 non-null	object
16	PaperlessBilling	7043 non-null	object
17	PaymentMethod	7043 non-null	object
18	MonthlyCharges	7043 non-null	float64
19	TotalCharges	7043 non-null	object

20 Churn 7043 non-null object
 dtypes: float64(1), int64(2), object(18)
 memory usage: 1.1+ MB
 None

Statistical Summary:

	SeniorCitizen	tenure	MonthlyCharges
count	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692
std	0.368612	24.559481	30.090047
min	0.000000	0.000000	18.250000
25%	0.000000	9.000000	35.500000
50%	0.000000	29.000000	70.350000
75%	0.000000	55.000000	89.850000
max	1.000000	72.000000	118.750000

Missing Values:

customerID	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0
tenure	0
PhoneService	0
MultipleLines	0
InternetService	0
OnlineSecurity	0
OnlineBackup	0
DeviceProtection	0
TechSupport	0
StreamingTV	0
StreamingMovies	0
Contract	0
PaperlessBilling	0
PaymentMethod	0
MonthlyCharges	0
TotalCharges	0
Churn	0

dtype: int64

Duplicate Values:

0

Missing Values After Preprocessing:

customerID	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0
tenure	0
PhoneService	0
MultipleLines	0
InternetService	0
OnlineSecurity	0
OnlineBackup	0
DeviceProtection	0
TechSupport	0
StreamingTV	0
StreamingMovies	0
Contract	0

```
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64
```

Columns After Removing Customer ID:

```
Index(['gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure',
      'PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity',
      'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV',
      'StreamingMovies', 'Contract', 'PaperlessBilling', 'PaymentMethod',
      'MonthlyCharges', 'TotalCharges', 'Churn'],
      dtype='object')
```

Encoded Dataset:

	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	\
0	0	0	1	0	1	0	
1	1	0	0	0	34	1	
2	1	0	0	0	2	1	
3	1	0	0	0	45	0	
4	0	0	0	0	2	1	

	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	\
0	1	0	0	2	
1	0	0	2	0	
2	0	0	2	2	
3	1	0	2	0	
4	0	1	0	0	

	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	\
0	0	0	0	0	0	
1	2	0	0	0	1	
2	0	0	0	0	0	
3	2	2	0	0	1	
4	0	0	0	0	0	

	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
0	1	2	29.85	29.85	0
1	0	3	56.95	1889.50	0
2	1	3	53.85	108.15	1
3	0	0	42.30	1840.75	0
4	1	2	70.70	151.65	1

Features:

```
Index(['gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure',
      'PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity',
      'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV',
      'StreamingMovies', 'Contract', 'PaperlessBilling', 'PaymentMethod',
      'MonthlyCharges', 'TotalCharges'],
      dtype='object')
```

Target:

Churn

Training Data Shape:

(5634, 19)

Testing Data Shape:

(1409, 19)

Decision Tree Model Trained Successfully.

Predicted Values:

[0 1 0 1 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0]

Model Evaluation:

Accuracy : 0.785

Precision: 0.6035

Recall : 0.5535

F1 Score : 0.5774

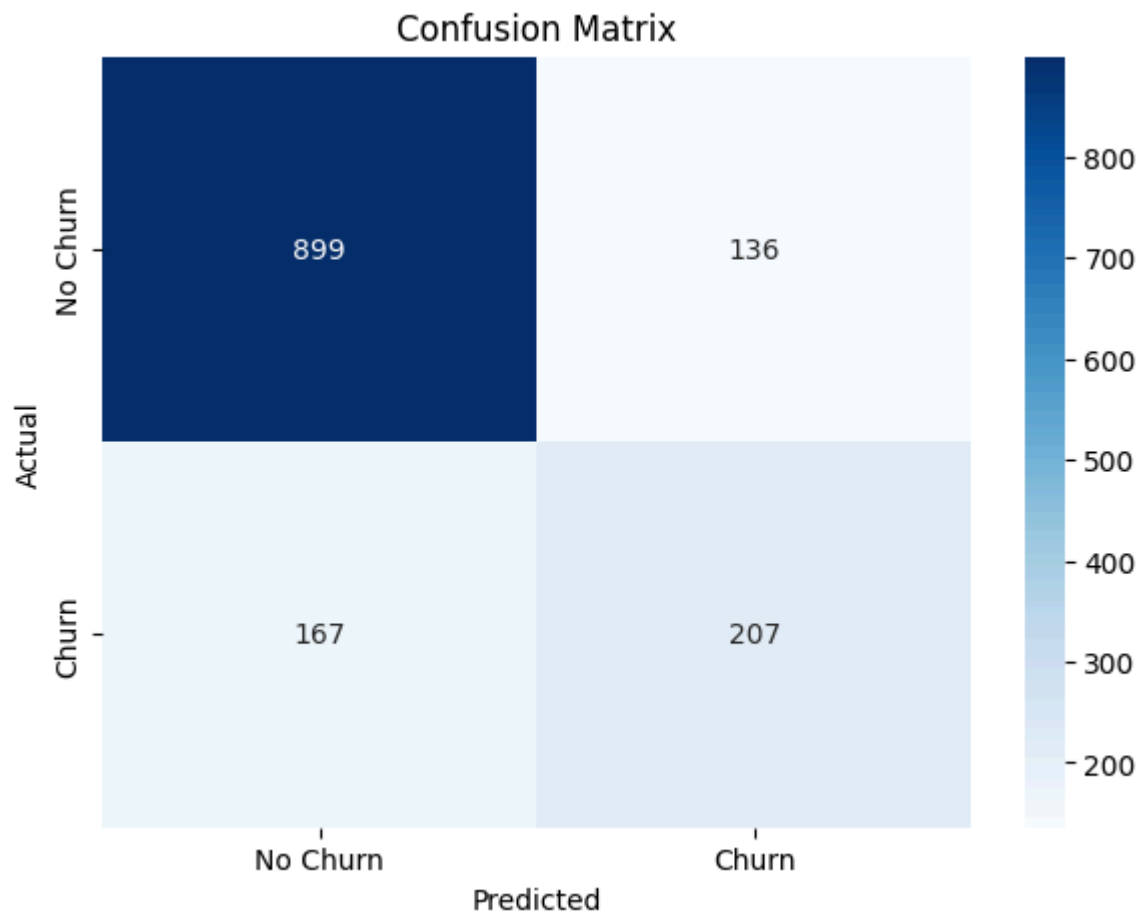
Classification Report:

	precision	recall	f1-score	support
0	0.84	0.87	0.86	1035
1	0.60	0.55	0.58	374
accuracy			0.78	1409
macro avg	0.72	0.71	0.72	1409
weighted avg	0.78	0.78	0.78	1409

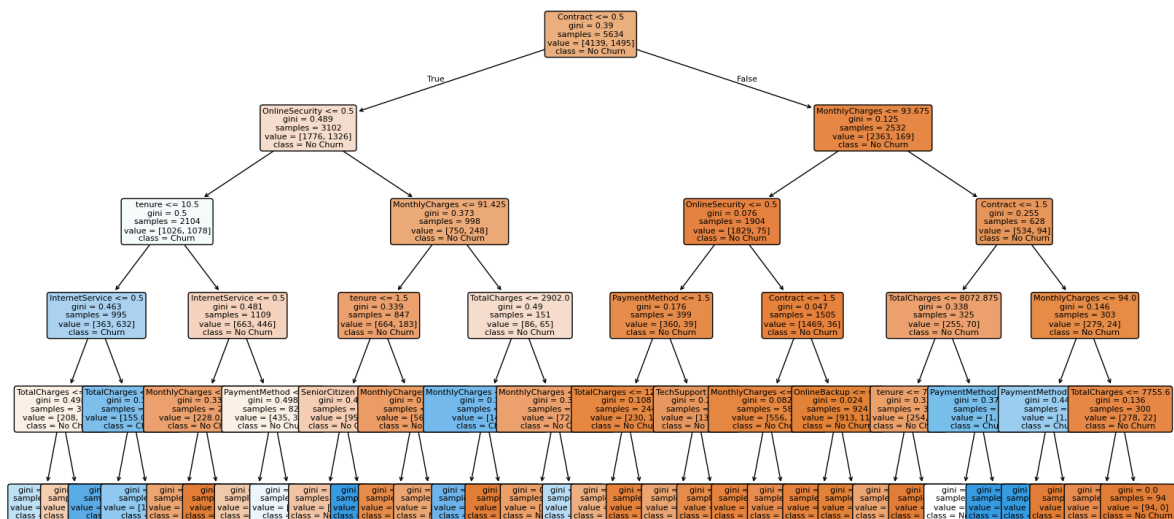
Confusion Matrix:

[[899 136]

[167 207]]



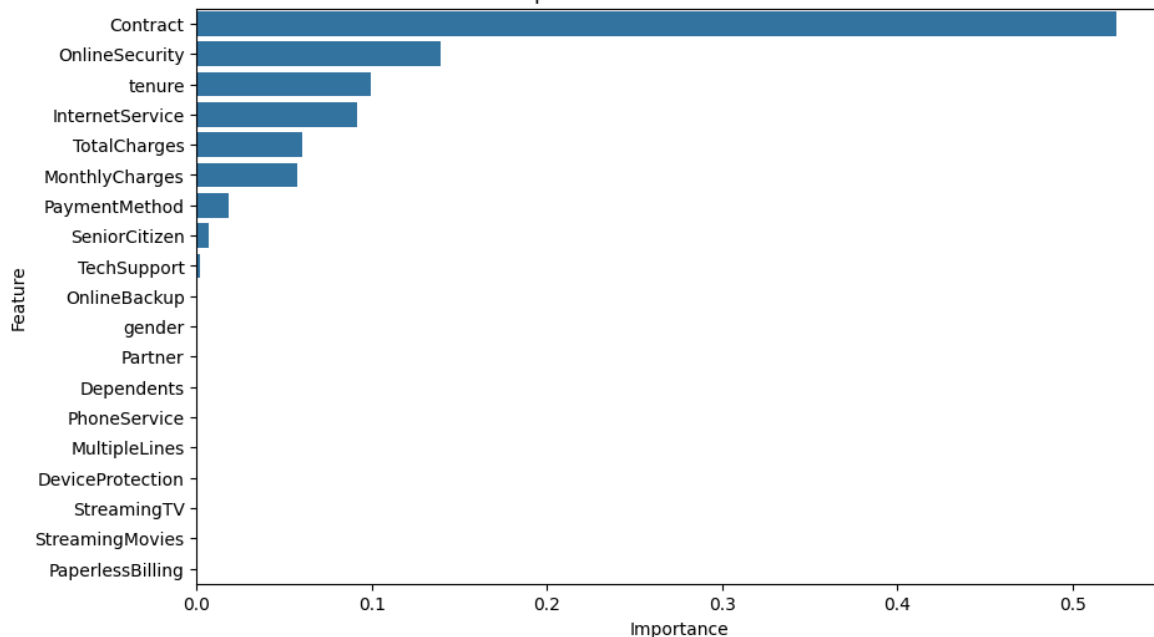
Decision Tree for Customer Churn Prediction



Feature Importance:

	Feature	Importance
14	Contract	0.525341
8	OnlineSecurity	0.138973
4	tenure	0.099174
7	InternetService	0.091187
18	TotalCharges	0.060525
17	MonthlyCharges	0.057434
16	PaymentMethod	0.018331
1	SeniorCitizen	0.006759
11	TechSupport	0.002142
9	OnlineBackup	0.000135
0	gender	0.000000
2	Partner	0.000000
3	Dependents	0.000000
5	PhoneService	0.000000
6	MultipleLines	0.000000
10	DeviceProtection	0.000000
12	StreamingTV	0.000000
13	StreamingMovies	0.000000
15	PaperlessBilling	0.000000

Feature Importance in Customer Churn Prediction



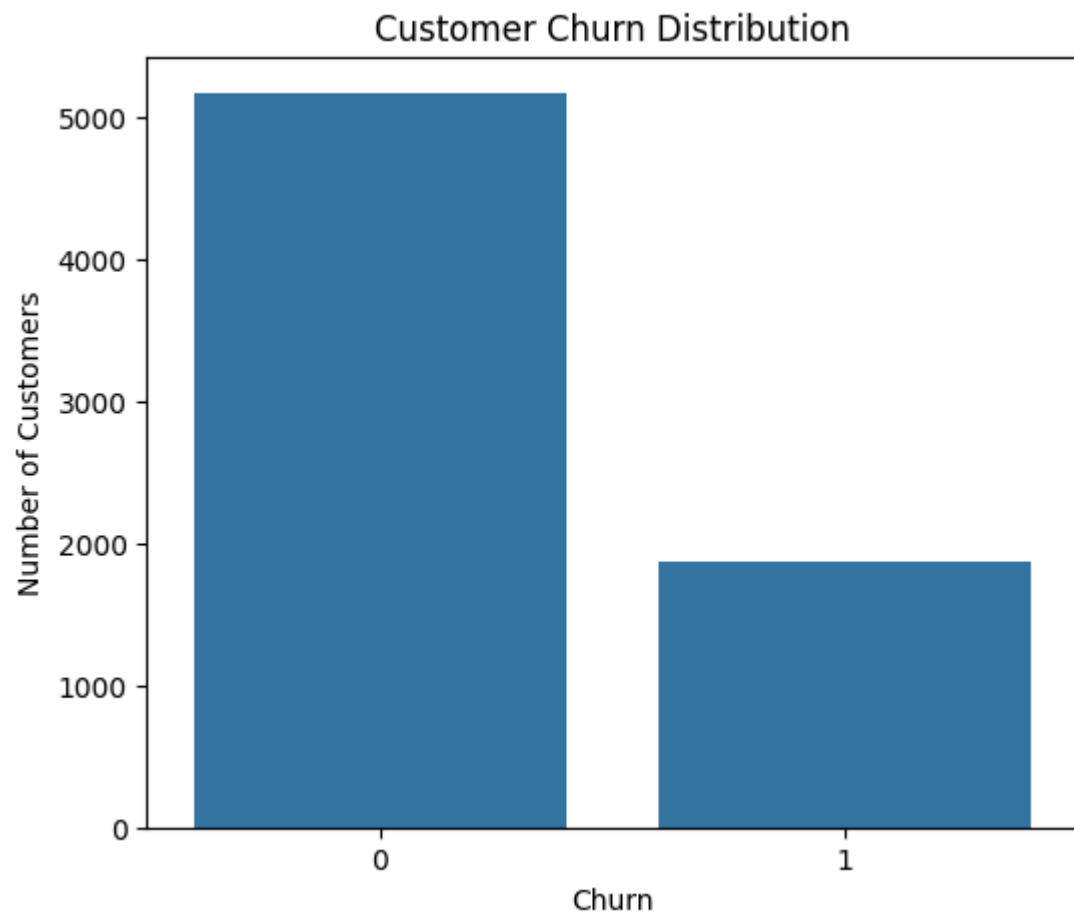
Customer Churn Distribution:

Churn

0 5174

1 1869

Name: count, dtype: int64



Customers Predicted to Churn:

	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	\
2280	0	1	0	0	8	1	
4460	1	0	1	0	18	1	
2136	1	0	0	1	15	1	
6529	1	1	0	0	2	1	
5389	0	0	1	0	51	1	
761	0	0	0	0	22	1	
5547	0	0	0	0	3	1	
1838	1	1	0	0	54	1	
6979	1	0	0	1	1	0	
346	0	0	0	0	2	1	
6633	0	0	0	0	1	1	
2519	1	0	1	0	2	1	
3510	0	0	0	0	8	1	
2246	0	0	1	1	1	1	
6304	0	0	0	0	9	1	
5300	0	0	1	1	1	1	
290	1	1	0	0	24	1	
642	1	0	0	0	1	1	
3856	1	0	1	0	61	1	
6314	0	0	0	0	6	1	

	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	...	\
2280	2	1	0	0	...	
4460	0	1	0	0	...	
2136	2	1	0	2	...	
6529	2	1	0	0	...	
5389	0	1	0	0	...	
761	0	1	0	0	...	
5547	0	1	0	0	...	
1838	2	1	0	2	...	
6979	1	0	0	0	...	
346	0	1	0	0	...	
6633	2	1	0	0	...	
2519	2	1	0	0	...	
3510	2	1	0	0	...	
2246	2	1	2	0	...	
6304	0	1	0	0	...	
5300	0	1	0	0	...	
290	2	1	0	0	...	
642	0	1	0	0	...	
3856	2	1	0	2	...	
6314	2	1	0	0	...	

	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	\
2280	2	2	2	0	1	
4460	2	0	0	0	0	
2136	0	2	0	0	1	
6529	0	0	0	0	1	
5389	0	0	0	0	0	
761	0	2	2	0	0	
5547	0	2	2	0	1	
1838	0	0	2	0	1	
6979	0	0	0	0	0	
346	0	2	2	0	1	
6633	0	0	0	0	1	
2519	0	2	0	0	1	
3510	0	0	0	0	1	
2246	0	2	2	0	1	

6304	0	2	0	0	1
5300	0	2	0	0	1
290	0	0	0	0	1
642	0	2	2	0	1
3856	2	2	2	0	1
6314	0	0	0	0	1

	PaymentMethod	MonthlyCharges	TotalCharges	Actual Churn	\
2280	1	100.15	908.55	0	
4460	2	78.20	1468.75	0	
2136	2	91.00	1430.05	0	
6529	3	74.85	156.40	1	
5389	3	69.15	3649.60	0	
761	2	89.25	1907.85	1	
5547	2	89.45	240.45	1	
1838	2	90.05	4931.80	0	
6979	3	24.20	24.20	0	
346	2	90.40	168.20	1	
6633	2	74.50	74.50	1	
2519	2	91.45	171.45	0	
3510	2	75.25	576.70	0	
2246	2	102.45	102.45	1	
6304	0	77.65	714.15	1	
5300	3	80.05	80.05	1	
290	2	76.10	1712.70	1	
642	2	89.55	89.55	1	
3856	2	114.10	7132.15	0	
6314	3	73.85	401.30	0	

	Predicted Churn
2280	1
4460	1
2136	1
6529	1
5389	1
761	1
5547	1
1838	1
6979	1
346	1
6633	1
2519	1
3510	1
2246	1
6304	1
5300	1
290	1
642	1
3856	1
6314	1

[20 rows x 21 columns]

Final Result:

Decision Tree Customer Churn Prediction completed successfully.

Model Accuracy: 78.50%

Precision: 60.35%

Recall: 55.35%

F1 Score: 57.74%

Questions and Answers

1. What is classification, and how does it differ from regression?

Classification is a supervised machine learning technique used to predict a **categorical class or label**.

Regression is used to predict a **continuous numerical value**.

- **Classification Example:** Predict whether a customer will **churn or not churn**.
- **Regression Example:** Predict the **price of a house**.

2. Explain the working principle of the Decision Tree algorithm.

A **Decision Tree** makes predictions by repeatedly splitting the dataset based on feature conditions.

The main steps are:

1. Select the best feature for splitting the data.
2. Divide the data into smaller groups.
3. Repeat the process for each group.
4. Stop when a suitable stopping condition is reached.
5. The final **leaf node** provides the predicted class.

The splits can be selected using criteria such as **Gini Index** or **Entropy**.

3. Differentiate between Gini Index and Entropy as splitting criteria.

Feature	Gini Index	Entropy
Purpose	Measures impurity	Measures uncertainty or impurity
Formula	$Gini = 1 - \sum p_i^2$	$Entropy = -\sum p_i \log_2(p_i)$
Calculation	Generally simpler and faster	Slightly more computationally expensive
Common Use	CART Decision Trees	Information Gain-based splitting

Both criteria help the Decision Tree select useful splits. A lower impurity indicates a more homogeneous group.

4. What is a Confusion Matrix? Explain its components.

A **Confusion Matrix** is a table used to evaluate the performance of a classification model.

Its main components are:

- **True Positive (TP):** Positive cases correctly predicted as positive.
- **True Negative (TN):** Negative cases correctly predicted as negative.
- **False Positive (FP):** Negative cases incorrectly predicted as positive.
- **False Negative (FN):** Positive cases incorrectly predicted as negative.

For customer churn prediction, a **False Negative** means a customer who actually churns was predicted as a non-churning customer.

5. Define Accuracy, Precision, Recall, and F1-Score. Why are these metrics important?

- **Accuracy:** Measures the proportion of total predictions that are correct.

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

- **Precision:** Measures how many predicted positive cases are actually positive.

$$\text{Precision} = TP / (TP + FP)$$

- **Recall:** Measures how many actual positive cases are correctly identified.

$$\text{Recall} = TP / (TP + FN)$$

- **F1-Score:** Combines Precision and Recall using their harmonic mean.

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

These metrics are important because accuracy alone may not adequately describe performance, especially when the classes are imbalanced.

6. What is overfitting in a Decision Tree? How can it be reduced?

Overfitting occurs when a Decision Tree becomes too complex and learns noise or very specific patterns from the training data. It may perform well on training data but poorly on unseen data.

Overfitting can be reduced by:

- Limiting the **maximum tree depth**.
- Increasing **minimum samples required for splitting**.
- Increasing **minimum samples required at leaf nodes**.
- Pruning unnecessary branches.
- Using ensemble methods such as Random Forest.

7. Why is customer churn prediction important for businesses?

Customer churn prediction helps businesses identify customers who may discontinue their service or stop purchasing.

It can help organizations:

- Identify potential churners.
- Understand factors associated with churn.
- Plan customer-retention activities.
- Improve customer satisfaction.
- Reduce potential revenue loss.
- Allocate retention resources effectively.

8. What are the advantages and limitations of the Decision Tree algorithm?

Advantages:

- Easy to understand and interpret.
- Requires relatively little data preprocessing.
- Can handle numerical and categorical features.
- Can model non-linear relationships.
- Provides useful feature-based decision rules.

Limitations:

- Can overfit easily.
- Small changes in data can produce a different tree.
- Deep trees can become complex.
- A single tree may have lower generalization performance than ensemble methods.
- It can be affected by imbalanced datasets.

9. Mention three real-world applications of Decision Tree Classification other than customer churn prediction.

1. **Medical Diagnosis** – Classifying patients based on symptoms and test results.
2. **Credit Risk Assessment** – Classifying loan applicants according to risk categories.
3. **Fraud Detection** – Classifying transactions as potentially fraudulent or legitimate.

10. How can the insights obtained from a churn prediction model help organizations improve customer retention and business profitability?

A churn model can help organizations identify **patterns and characteristics associated with customer churn**. Businesses can use these insights to:

- Identify customers who may require attention.
- Understand common factors associated with churn.
- Improve products or services.
- Provide relevant offers or support.
- Monitor customer satisfaction.
- Develop targeted retention programs.
- Reduce customer loss and associated revenue impact.

The model should be used as a decision-support tool and evaluated regularly to ensure that its predictions remain reliable.